

CS 6341 Robotics Homework 1

Instructor Yu Xiang

September 3, 2026

In this homework, write down your solutions for problem 1 and finish the coding problem 2. Upload your solutions and code to eLearning. Our TA will check your solutions and run your scripts to verify them.

Problem 1

(3 points)

Grübler's Formula. Exercise 2.7 in Lynch and Park, Modern Robotics.

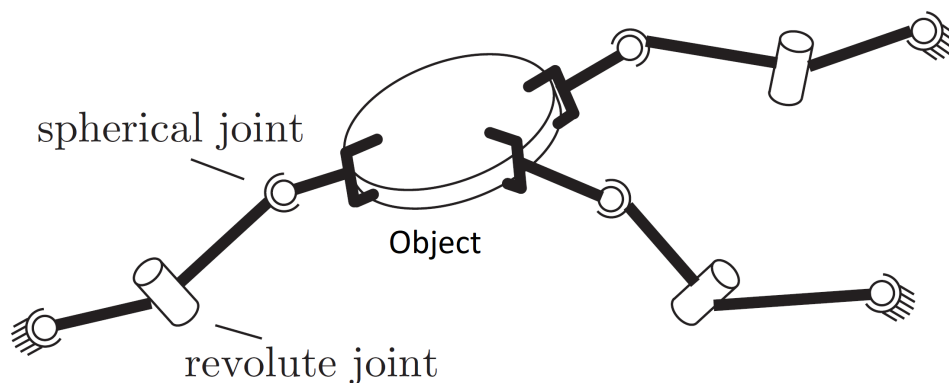


Figure 1: Three cooperating SRS arms grasping a common object.

Three identical SRS open-chain arms are grasping a common object, as shown in Figure 1. Find the number of degrees of freedom of this system using Grübler's formula. Write down how you apply Grübler's formula to this system.

Problem 2

(7 points)

ROS 2 programming and RViz.

In this problem, you will learn how to create and build a ROS 2 workspace and visualize a Fetch mobile manipulator and a SO101 arm in RViz. We will use **ROS 2 Jazzy Jalisco**. You can either install ROS 2 Jazzy directly on Ubuntu 24.04 or use the Docker environment introduced in the lecture.

Refer to the ROS 2 Jazzy documentation if needed:

<https://docs.ros.org/en/jazzy/>.

(2.1) Set up and test your ROS 2 environment.

For Ubuntu users: Install ROS 2 Jazzy directly by following:

<https://docs.ros.org/en/jazzy/Installation/Ubuntu-Install-Debs.html>

After installation, source ROS 2 and test RViz:

- `source /opt/ros/jazzy/setup.bash`
- `rviz2`

For Windows and Mac users: Use the ROS 2 Jazzy Docker image introduced in the lecture:

<https://github.com/Tiryoh/docker-ros2-desktop-vnc>

Create a data folder on your host computer. For example:

- Windows: `C:\data`
- Mac: `~/data`

When creating the Docker container, mount this folder to `/home/ubuntu/data` inside Docker using the `-v` option.

For Windows PowerShell:

```
docker run -p 6080:80 --shm-size=512m --name ros2-jazzy -v C:\data:/home/ubuntu/data ghcr.io/tiryoh/ros2-desktop-vnc:jazzy
```

For Mac Terminal:

```
docker run -p 6080:80 --shm-size=512m --name ros2-jazzy -v ~/data:/home/ubuntu/data ghcr.io/tiryoh/ros2-desktop-vnc:jazzy
```

After the Docker container starts, open:

<http://127.0.0.1:6080/>

in your web browser. You should see the Ubuntu desktop.

Open a terminal inside the Ubuntu desktop and run:

- `source /opt/ros/jazzy/setup.bash`
- `rviz2`

You should see the RViz interface.

(2.2) Create a ROS 2 workspace.

A ROS 2 workspace is a directory used to organize and build your own ROS 2 packages. Refer to the following tutorial if needed:

<https://docs.ros.org/en/jazzy/Tutorials/Beginner-Client-Libraries/Creating-A-Workspace/Creating-A-Workspace.html>

For this homework, create the workspace under your data directory:

```
mkdir -p ~/data/ros2_ws/src
```

Your ROS 2 workspace should therefore be:

```
~/data/ros2_ws
```

and ROS 2 packages should be placed under:

```
~/data/ros2_ws/src
```

For Docker users, because `/data` is mounted from the host computer, your source code will also be stored on your Windows or Mac computer.

(2.3) Add packages to your ROS 2 workspace.

We are going to use two existing ROS 2 packages to visualize the Fetch mobile manipulator.

First, go to the `src` directory:

```
cd ~/data/ros2_ws/src
```

Clone the following repositories:

- `fetch_description`:
`git clone https://github.com/IRVLUTD/fetch_ros2_IRVL.git`
- `urdf_tutorial`:
`git clone -b ros2 https://github.com/ros/urdf_tutorial.git`

You may also need to install the `urdf_launch` package:

```
sudo apt update
sudo apt install ros-jazzy-urdf-launch
```

You can read more about ROS 2 packages here:

<https://docs.ros.org/en/jazzy/Tutorials/Beginner-Client-Libraries/Creating-Your-First-ROS2-Package.html>

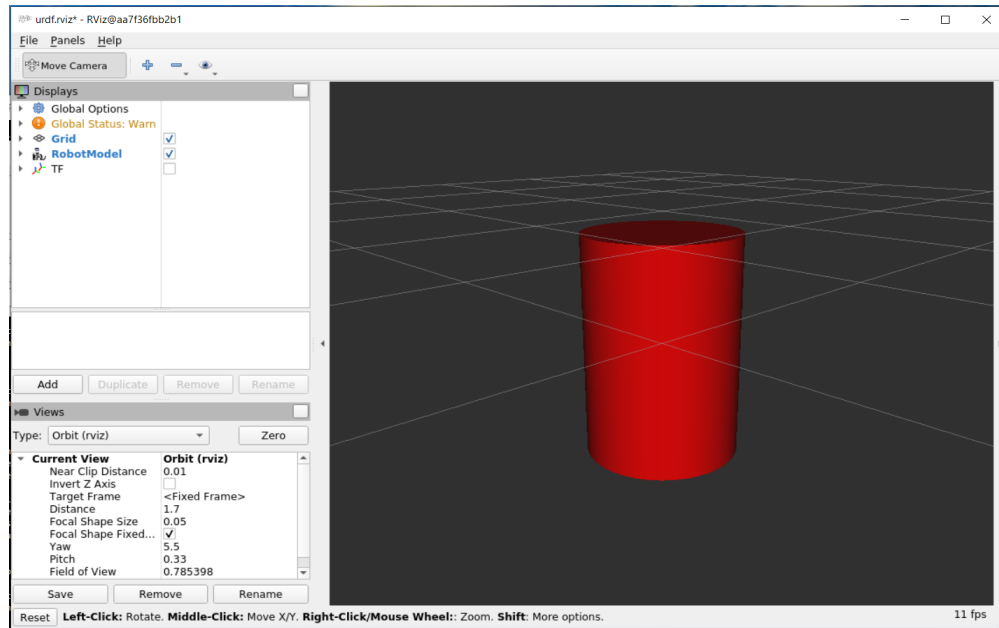


Figure 2: RViz interface after running `ros2 launch urdf_tutorial display.launch.py`.

(2.4) Build and test your ROS 2 workspace.

Before building the workspace, make sure ROS 2 Jazzy has been sourced:

```
source /opt/ros/jazzy/setup.bash
```

Go to the root of your workspace:

```
cd ~/data/ros2_ws
```

Build the workspace:

```
colcon build
```

Do not use `sudo colcon build`.

If the workspace has been built successfully, source your workspace:

```
source install/setup.bash
```

Then test the `urdf_tutorial` package:

```
ros2 launch urdf_tutorial display.launch.py
```

The command launches the `display.launch.py` file in the `urdf_tutorial` package. You should see the RViz interface as shown in Figure 2.

(2.5) Visualize the Fetch robot model.

Create a new launch file named `display_fetch.launch.py` in:

```
urdf_tutorial/launch
```

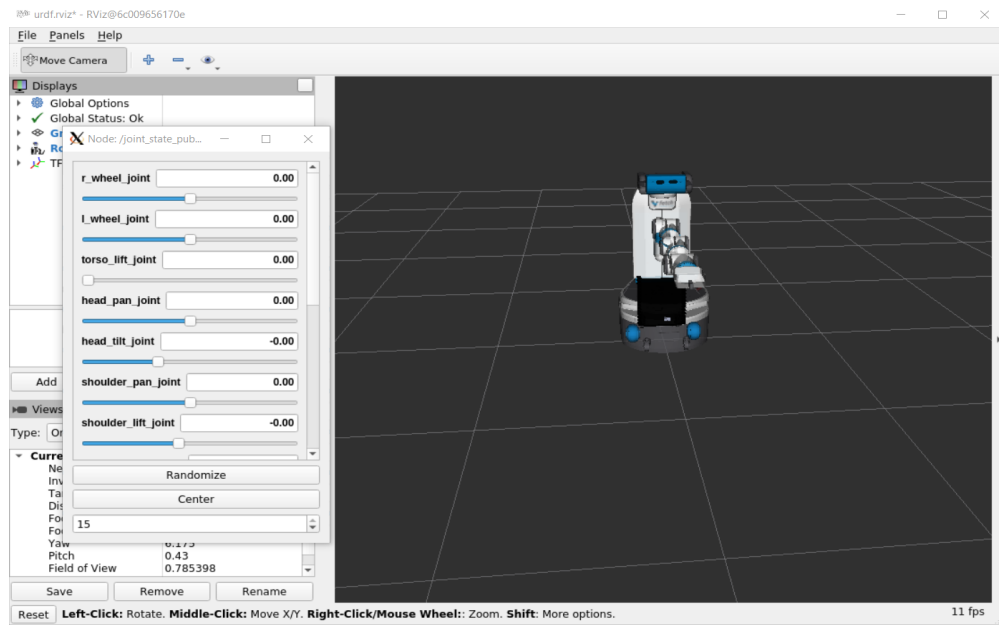


Figure 3: RViz visualization of a Fetch mobile manipulator.

by copying the existing `display.launch.py` file.

Modify the new launch file so that it displays the Fetch robot instead of the default URDF model.

The Fetch robot definition is located at:

```
fetch_ros2_IRVL/fetch_description/robots/fetch.urdf
```

After modifying the launch file, rebuild your ROS 2 workspace:

```
cd ~/data/ros2_ws
colcon build
source install/setup.bash
```

Then run:

```
ros2 launch urdf_tutorial display_fetch.launch.py
```

You should see the Fetch mobile manipulator in RViz as shown in Figure 3.

(2.6) Tune the robot joints.

Use the `/joint_state_publisher` window to set the following robot arm joints to the corresponding values:

- `shoulder_pan_joint`: -1.32
- `shoulder_lift_joint`: 0.7
- `upperarm_roll_joint`: 0.0
- `elbow_flex_joint`: -2.01

- forearm_roll_joint: 0.0
- wrist_flex_joint: -0.56
- wrist_roll_joint: 0.0

You can keep the other joint values at their default values.

Capture a screenshot showing the Fetch robot in RViz with the specified joint configuration and include it in your homework submission.

(2.7) Visualizing the SO-101 robot arm.

In addition to the Fetch mobile manipulator, we will also visualize the SO-101 robot arm.

The ROS 2 description package for the SO-101 is available at:

https://github.com/IRVLUTD/so101_description

Using what you learned in the previous parts of this problem, add the so101_description package to your ROS 2 workspace, rebuild the workspace, and source it appropriately. Then figure out how to launch the SO-101 visualization in RViz using the files provided in the package.

Once the SO-101 is displayed in RViz, use the Joint State Publisher window to set the following joint values:

- shoulder_pan: 0.384
- shoulder_lift: -0.632
- elbow_flex: 1.014
- wrist_flex: -0.887
- wrist_roll: 1.453
- gripper: 0.863

Adjust the RViz camera if needed so that the robot is clearly visible. Capture a screenshot showing both the SO-101 robot in RViz and the Joint State Publisher window with the specified joint values. Include the screenshot in your homework submission.

Submission guideline: Zip your src folder in your workspace and submit the zip file to eLearning. Make sure you have screen captures included for (2.6 and 2.7). Our TA will build your workspace and run commands to verify it.